

CCP Cheat Sheet

IAM (Identity and Access Management)

- Users — Mapped to a physical user, has a password for AWS Console
- Groups — Contains users only
- Policies — JSON document that outlines permissions for users or groups
- Roles — For EC2 instances or AWS services
- Security — MFA + Password Policy
- AWS CLI — Manage your AWS services using the command-line *Interface*
- AWS SDK — Manage your AWS services using a programming language (*Software Development Kit*)
- Access Keys — Access AWS using the CLI or SDK
- Audit — IAM Credential Reports & IAM Access Advisor

EC2 (Elastic Compute Cloud)

- EC2 Instance — AMI(OS) + Instance Size (CPU + RAM) + Storage + Security Group + EC2 User Data
- Security Groups — Firewall attached to the EC2 instance
- EC2 User Data — Script launched at the first start of an instance
↳ *Bootstrapping*
- SSH — Start terminal into our EC2 Instances (port 22)
- EC2 Instance Role — Link to IAM roles
- Purchasing Options — On-demand, Spot, Reserved (Standard + Convertible), Dedicated Host, Dedicated Instance, *Savings Plan*

EC2 Instance Storage

- EBS Volumes
 - Network drives attached to one EC2 instance at a time
 - Mapped to an Availability Zones
 - Can use EBS Snapshots for backups/transferring EBS volumes across AZ
- AMI — Create ready-to-use EC2 instance with our customizations
- EC2 Image Builder — Automatically build, test, and distribute AMIs

- EC2 Instance Store
 - High-performance hardware disk attached to our EC2 instance
 - Lost if our instance is stopped/terminated
- EFS — Network file system, can be attached to 100s of instances in a region
- EFS-IA — Cost-optimized storage class of infrequent access file
- FSx for Windows — Network File System for Windows servers
- FSx for Lustre — High-Performance Computing Linux file system

ELB & ASG

↳ (Linux + Cluster)

- High Availability vs Scalability (vertical and horizontal) vs Elasticity vs Agility in the Cloud
- Elastic Load Balancers (ELB)
 - Distribute traffic across backend EC2 instances, can be Multi-AZ
 - Supports health checks
 - 4 types: Classic (old), Application (HTTP-L7), Network (TCP-L4), Gateway (L3)
- Auto Scaling Groups (ASG)
 - Implement Elasticity for your application, across multiple AZ
 - Scale EC2 instances based on the demand on your system, and replace unhealthy instances
 - Integrated with the ELB

ELB: Elastic Load Balancing

ASG: Auto Scaling Group

Amazon S3

- Buckets vs Objects — global unique name, tied to a region
- S3 Security — IAM policy, S3 Bucket Policy (public access), S3 Encryption
- S3 Websites — Host a static website on Amazon S3
- S3 Versioning — Multiple versions for files, prevent accidental deletes
- S3 Replication — Same-region or cross-region, must enable versioning

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- **S3 Storage Classes** — Standard, IA, 1Z-IA, Intelligent, Glacier (Instant, Flexible, Deep)
 - **Snow Family** — Import data onto S3 through a physical device, edge computing
 - **SnowCone & SnowCone SSD** — Storage capacity → 8TB HDD, 14TB SSD, Storage capacity → 8TBsize up to 24TB, online and offline
 - **Snowball Edge & Storage Optimized** — Storage capacity → 80TB usable, Storage capacity → up to Petabytes, offline
 - **SnowMobile** — Storage capacity → < 100PB, Storage capacity → up to Exabytes, offline
 - **OpsHub** — Desktop application to manage Snow Family devices
 - **Storage Gateway** — Hybrid solution to extend on-premises storage to S3 or cloud as S3 is a part of cloud
- IA: Infrequency Access

Performance across the S3 Storage Classes

	S3 Standard	S3 Intelligent-Tiering*	S3 Standard-IA	S3 One Zone-IA†	S3 Glacier Instant Retrieval	S3 Glacier Flexible Retrieval	S3 Glacier Deep Archive
Designed for durability	99.999999999% (11 9's)	99.999999999% (11 9's)	99.999999999% (11 9's)	99.999999999% (11 9's)	99.999999999% (11 9's)	99.999999999% (11 9's)	99.999999999% (11 9's)
Designed for availability	99.99%	99.9%	99.9%	99.5%	99.9%	99.99%	99.99%
Availability SLA	99.9%	99%	99%	99%	99%	99%	99.9%
Availability Zones	≥3	≥3	≥3	1	≥3	≥3	≥3
Minimum capacity charge per object	N/A	N/A	128 KB	128 KB	128 KB	40 KB	40 KB
Minimum storage duration charge	N/A	N/A	30 days	30 days	90 days	90 days	180 days
Retrieval charge	N/A	N/A	per GB retrieved	per GB retrieved	per GB retrieved	per GB retrieved	per GB retrieved
First byte latency	milliseconds	milliseconds	milliseconds	milliseconds	milliseconds	minutes or hours	hours
Storage type	Object	Object	Object	Object	Object	Object	Object
Lifecycle transitions	Yes	Yes	Yes	Yes	Yes	Yes	Yes

via — <https://aws.amazon.com/s3/storage-classes/>

Databases & Analytics

- **Relational Databases-OLTP** — RDS & Aurora (SQL)

- **Differences between Multi-AZ, Read Replicas, Multi-Region**

- **In-memory Database** — ElasticCache

- **Key/Value Database** — DynamoDB (serverless) & DAX (cache for DynamoDB)

For Relational Database
Also for RDB but better, faster and serverless
Improves app performance
millisecond response, lets you write schemas

- Warehouse-OLAP — Redshift (SQL)
- Hadoop Cluster — EMR
- Athena — Query data on Amazon S3 (serverless & SQL)
- QuickSight — Dashboard on your data (serverless)
- DocumentDB — “Aurora for MongoDB” (JSON- NoSQL database)
- Amazon QLDB — Financial Transactions Ledger (immutable journal, cryptographically verifiable)
- Amazon Managed Blockchain — managed Hyperledger Fabric & Ethereum blockchains
- Glue — Managed ETL (Extract Transform Load) and Data Catalog service
- Database Migration — DMS
- Neptune — Graph Database (Social Network)

OLAP: Online Analytics Processing

Read replicas, Multi-AZ deployments, and multi-region deployments

Amazon RDS read replicas complement Multi-AZ deployments. While both features maintain a second copy of your data, there are differences between the two:

Multi-AZ deployments	Multi-Region deployments	Read replicas
Main purpose is high availability	Main purpose is disaster recovery and local performance	Main purpose is scalability
Non-Aurora: synchronous replication; Aurora: asynchronous replication	Asynchronous replication	Asynchronous replication
Non-Aurora: only the primary instance is active; Aurora: all instances are active	All regions are accessible and can be used for reads	All read replicas are accessible and can be used for readscaling
Non-Aurora: automated backups are taken from standby; Aurora: automated backups are taken from shared storage layer	Automated backups can be taken in each region	No backups configured by default
Always span at least two Availability Zones within a single region	Each region can have a Multi-AZ deployment	Can be within an Availability Zone, Cross-AZ, or Cross-Region
Non-Aurora: database engine version upgrades happen on primary; Aurora: all instances are updated together	Non-Aurora: database engine version upgrade is independent in each region; Aurora: all instances are updated together	Non-Aurora: database engine version upgrade is independent from source instance; Aurora: all instances are updated together
Automatic failover to standby (non-Aurora) or read replica (Aurora) when a problem is detected	Aurora allows promotion of a secondary region to be the master	Can be manually promoted to a standalone database instance (non-Aurora) or to be the primary instance (Aurora)

Other Compute

- Docker — Container technology to run applications

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- ECS (Elastic Container Service)— Run docker containers on EC2 instance
 - Fargate
 - Run Docker containers without provisioning the infrastructure
 - Serverless offering (no EC2 instances)
 - ECR (Elastic Container Registry) — Private Docker Image Repository
 - Batch — Run batch jobs on AWS across managed EC2 instance
 - Lightsail — Predictable & low pricing for simple applications & DB

Lambda

- Lambda — Serverless, Function as a Service, seamless scaling, reactive
- Lambda Billings
 - By the time run x by the RAM Provisioned
 - By the number of invocations
- Language Support — Many programming languages except (arbitrary) Docker
- Invocation time — up to 15 minutes
- Use cases
 - Create Thumbnails for images uploaded onto S3
 - Run a Serverless cron job
- API Gateway — Expose Lambda functions as HTTP API

Deployment

- CloudFormation (AWS only)
 - Infrastructure as Code, works with almost all AWS resources
 - Repeat across Regions & accounts
- Beanstalk (AWS only)
 - Platform as a Service (PaaS), limited to certain programming languages or Docker
 - Deploy code consistently with a known architecture: ex, ALB+BC+RDS
- CodeDeploy (Hybrid) — Deploy & upgrade any application onto servers

- **Systems Manager (Hybrid)** — Patch, configure, and run commands at scale
- **OpsWorks (Hybrid)** — Managed Chef and Puppet in AWS
- **CodeCommit** — Store code in private git repository (version controlled)
- **CodeBuild** — Build & Test code in AWS
- **CodeDeploy** — Deploy code onto servers
- **CodePipeline** — Orchestration of pipeline (from code to build to deploy)
- **CodeArtifact** — Store software packages/dependencies on AWS
- **CodeStar** — Unified view for allowing developers to do CI/CD and code
- **Cloud9** — Cloud IDE with collab
- **AWS CDK (Cloud Development Kit)** — Defined your cloud infrastructure using a programming language

Leveraging the AWS Global Application

- **Global DNS: Route 53**
 - It is great to route users to the closet deployment with the least latency
 - Great for disaster recovery strategies
- **Global Content Delivery Network (CDN): CloudFront**
 - Replicate part of your application to AWS Edge Locations — decrease latency
 - Cache common requests — improved user experience and decreased latency
- **S3 Transfer Acceleration**
 - Accelerate global uploads & download into Amazon S3
- **AWS Global Accelerator**
 - Improve global application availability and performance using the AWS global network

- **AWS Outposts**
 - Deploy Outposts Racks in your own Data Centers to extend AWS services
- **AWS WaveLenth**
 - Brings AWS services to the edge of the 5G networks
 - Ultra-low latency applications
- **AWS Local Zones**
 - Brings AWS resources (compute, database, storage, ...) closer to your users
 - Good for latency-sensitive applications

Cloud Integration

- **SQS (Simple Queue Service)**
 - Queue service in AWS
 - Multiple Producers, messages are kept up to 14 days
 - Multiple Consumers share the read and delete message
 - Used to **decouple** applications in AWS
- **SNS (Simple Notification Service)**
 - Notification service in AWS
 - Subscribers: Email, Lambda, SQS, HTTP, Mobile, ...
 - Multiple Subscribers, send all messages to all of them
 - No message retention
- **Kinesis** — Real-time data streaming, persistence, and analysis
- **Amazon MQ** — Managed message broker for ActiveMQ and RabbitMQ in the cloud (MQTT, AMQP...protocols)

Cloud Monitoring

- **CloudWatch**
 - **Metrics** — Monitor the performance of AWS services and billing metrics
 - **Alarms** — Automate notification, perform EC2 action, notify to SNS based on metrics
 - **Logs** — Collect log files from EC2 instances, servers, Lambda functions...
 - **Events (or EventBridge)** — React to events in AWS, or trigger a rule on a schedule

- **CloudTrail** — Audit API calls made within your AWS account
- **CloudTrail Insights** — Automated analysis of your CloudTrail Events
- **X-Ray** — Trace requests made through your distributed applications
- **AWS Health Dashboard** — Status of all AWS services across all regions
- **AWS Account Health Dashboard** — AWS events that impact your infrastructure
- **Amazon CodeGuru** — Automated code reviews and application performance recommendations

AWS Security & Compliance

- **Share Responsibility on AWS**
- **Shield** — Automatic DDoS Protection + 24/7 support for advanced
- **WAF** — Firewall to filter incoming requests based on rules
- **KMS** — Encryption keys managed by AWS
- **CloudHSM** — Hardware encryption, we manage encryption keys
- **AWS Certificate Manager** — Provision, manage, and deploy SSL/TLS certificates
- **Artifact** — Get access to compliance reports such as PCI, ISO, etc ...
- **GuardDuty** — Find malicious behavior with VPC, DNS & CloudTrail Logs
- **Inspector** — Find software vulnerabilities in EC2, ECR images, and Lambda functions
- **Network Firewall** — Protect VPC against network attacks
- **Config** — Track config changes and compliance against rules
- **Macie** — Find sensitive data (e.g. PII data) in Amazon S3 buckets
- **CloudTrail** — Track API calls made by users within an account
- **Amazon Detective** — Find the root cause of security issues or suspicious activities

- **AWS Abuse** — Report AWS resources used for abusive or illegal purposes
- **Root user privileges**
 - Change account setting
 - Close your AWS account
 - Change or cancel your AWS Support plan
 - Register as a seller in the Reserved Instance Marketplace

AWS Machine Learning

- **Rekognition** — Face detection, labeling, celebrity recognition
- **Transcribe** — Audio to Text (e.g. subtitles)
- **Polly** — Text to Audio
- **Translate** — Translations
- **Lex** — Build conversation bots (e.g. chatbots), (like Alexa)
- **Connect** — Cloud contact center
- **Comprehend** — Natural Language Processing
- **SageMaker** — Machine Learning for every developer and data scientist
- **Forecast** — Build highly accurate forecasts
- **Kendra** — ML-powered search engine
- **Personalize** — Real-time personalized recommendations
- **Textract** — Detect text and data in documents

AWS VPC & Network

- **VPC** — Virtual Private Cloud
- **Subnets** — Tied to an AZ, network partition of the VPC
- **Internet Gateway** — At the VPC level, provide Internet Access
- **NAT Gateway / Instances** — Give internet access to private subnets
- **NACL** — Stateless, subnet rules for inbound and outbound
- **Security Groups** — Stateful, operate at the EC2 instance level or ENI

- **VPC Peering** — Connect two VPC with non-overlapping IP ranges, nontransitive
- **Elastic IP** — Fixed public IPv4, ongoing cost if not in-use
- **VPC Endpoints** — Provide Private access to AWS Services within VPC
- **PrivateLink** — Privately connect to a service in a 3rd party VPC
- **VPC Flow Logs** — network traffic logs
- **Site to Site VPN** — VPN over public internet between op-premises DC and AWS
- **Client VPN** — OpenVPN connection from your computer into your VPC
- **Direct Connect** — Direct private connection to AWS
- **Transit Gateway** — Connect thousands of VPC and on-premises networks together

Account Best Practices

- Operate multiple accounts using **Organizations**
- Use **SCP** (service control policies) to restrict account power
- Easily setup multiple accounts with best-practices with **AWS Control Tower**
- Use **Tags & Cost Allocation Tags** for easy management & billing
- **IAM guidelines** — MFA, least-privilege, password policy, password rotation
- **Config** To record all resources configurations & compliance over time
- **CloudFormation** to deploy stack across accounts and regions
- **Trusted Advisor** to get insights, Support Plan adapted to your needs
- **If your Account is compromised** — change the root password, delete and rotate all passwords/keys, contact the AWS support
- Allow users to create pre-defined stacks defined by admins using **AWS Service Catalog**

Billing and Costing Tools

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- **Compute Optimizer** — Recommends resources' configuration to reduce cost
 - **Pricing Calculator** — Cost of services on AWS
 - **Billing Dashboard** — High-level overview + free tier dashboard
 - **Cost Allocation Tags** — Tag resources to create detailed reports
 - **Cost and Usage Reports** — The most comprehensive billing dataset
 - **Cost Explorer** — View current usage (detailed) and forecast usage
 - **Billing Alarms** — in us-east-1 — track overall and per-service billing
 - **Budgets** — More advanced — track usage, costs, RI, and get alerts
 - **Saving Plans** — Easy way to save based on long-term usage of AWS
 - **Cost Anomaly Detection** — Detect unusual spends using Machine Learning
 - **Service Quotas** — Notify you when you're close to the service quota threshold

Advanced Identity

- **IAM**
 - Identity and Access Management inside your AWS account
 - For users that you trust and belong to your company
- **Organizations** — Manage multiple accounts
- **Security Token Service (STS)** — temporary, limited-privileges credentials to access AWS resources
- **Cognito** — Create a database of users for your mobile & web applications
- **Directory Services** — Integrate Microsoft Active Directory in AWS
- **IAM Identity Center** — One login for multiple AWS accounts & applications

AWS Support Plan

There are four AWS support plans available:

- **Basic** — billing and account support only (access to forums only).

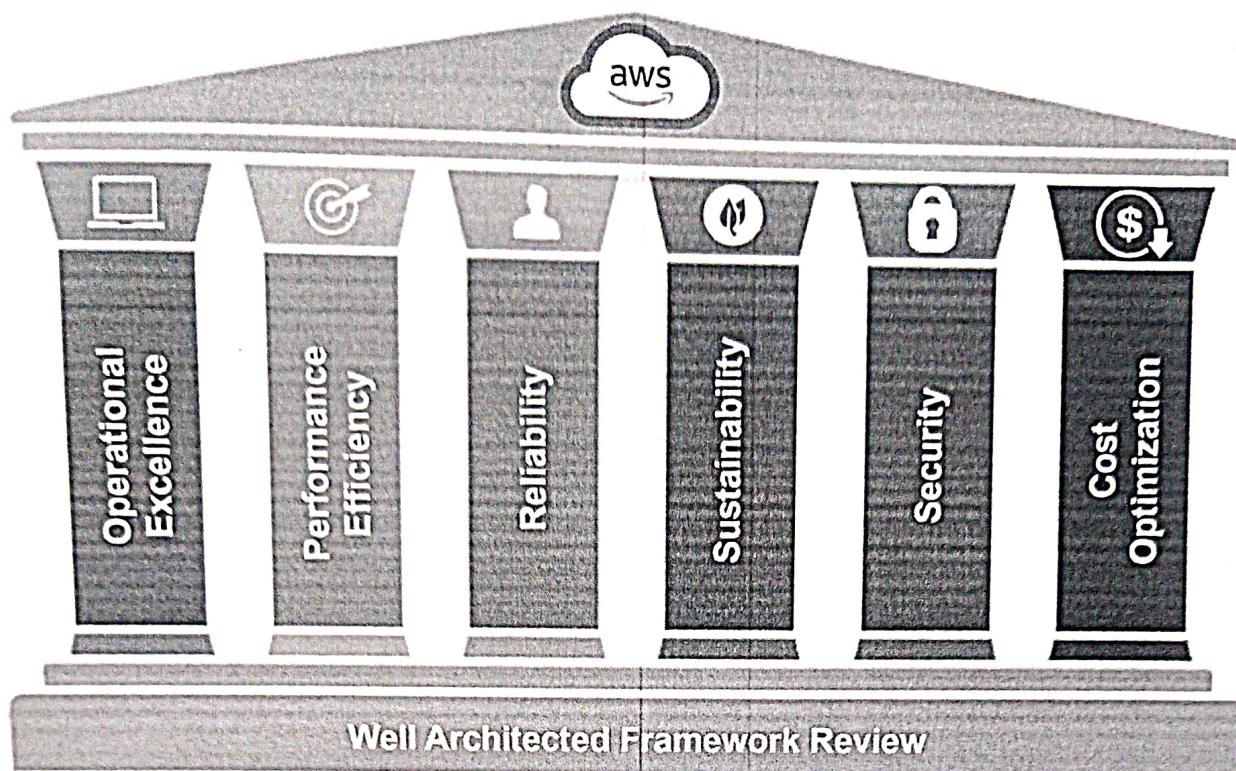
- **Developer** — business hours support via email.
- **Business** — 24x7 email, chat, and phone support.
- **Enterprise** — 24x7 email, chat, and phone support.

	Developer	Business	Enterprise
AWS Trusted Advisor Best Practice Checks	7 Core checks	Full set of checks	Full set of checks
Enhanced Technical Support	• Business hours email access to Cloud Support Associates • Unlimited cases / 1 primary contact	• 24x7 phone, email, and chat access to Cloud Support Engineers • Unlimited cases / unlimited contacts (IAM supported)	• 24x7 phone, email, and chat access to Cloud Support Engineers • Unlimited cases / unlimited contacts (IAM supported)
Case Severity / Response Times*	• General guidance: < 24 hours** • System Impaired: < 12 hours**	• General guidance: < 24 hours • System Impaired: < 12 hours • Production system impaired: < 4 hours • Production system down: < 1 hour	• General guidance: < 24 hours • System impaired: < 12 hours • Production system impaired: < 4 hours • Production system down: < 1 hour • Business-critical system down: < 15 minutes

	Developer	Business	Enterprise
Architectural Guidance	General	Contextual to your use-cases	Consultative review and guidance based on your applications
Programmatic Case Management		AWS Support API	AWS Support API
Proactive Programs and Services		Access to Infrastructure Event Management for additional fee	• Infrastructure Event Management • Well-Architected Reviews • Access to proactive reviews, workshops, and deep dives
Technical Account Management			Designated Technical Account Manager (TAM) to proactively monitor your environment and assist with optimization and coordinate access to programs and AWS experts
Account Assistance			Concierge Support Team

AWS Well-Architected and the Six Pillars

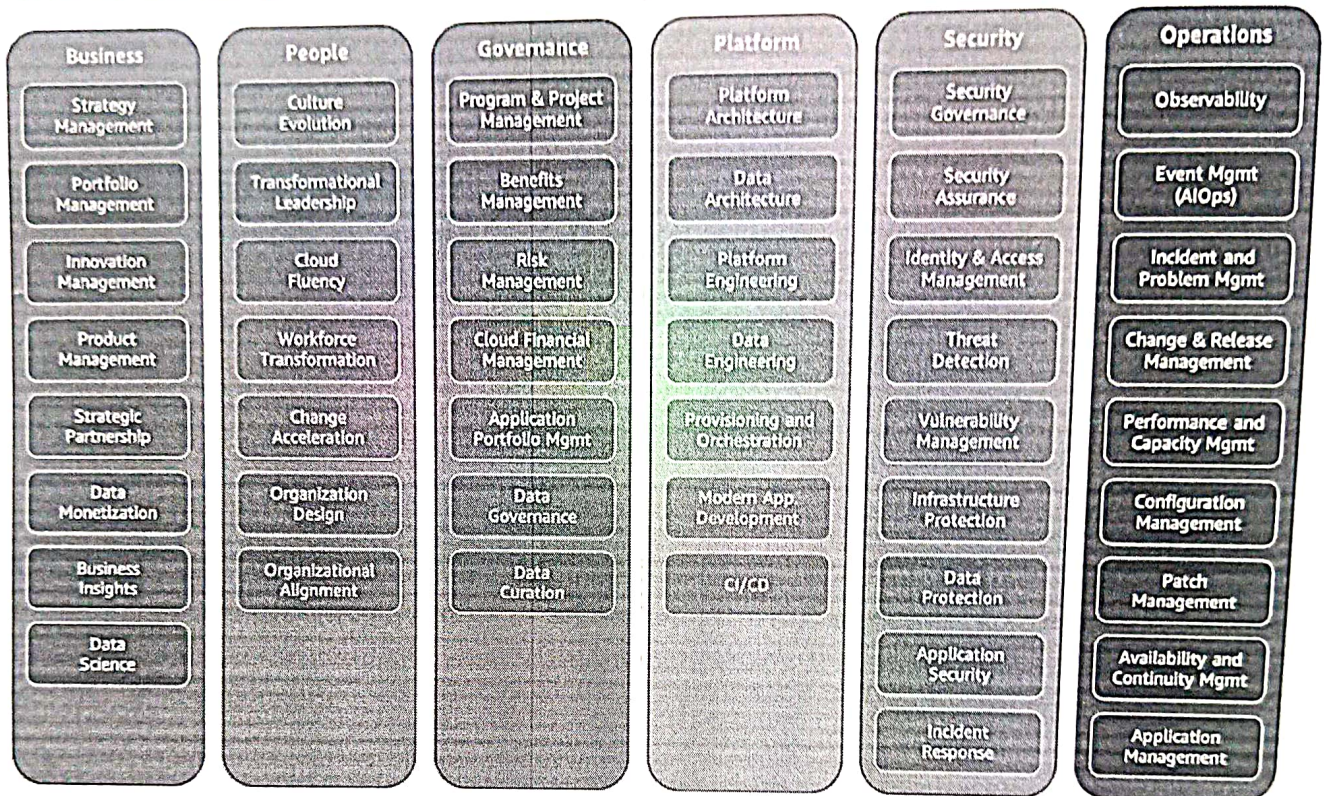
The AWS Well-Architected Framework describes key concepts, design principles, and architectural best practices for designing and running workloads in the cloud. By answering a few foundational questions, learn how well your architecture aligns with cloud best practices and gain guidance for making improvements.



- **Operational Excellence Pillar** — Includes the ability to run and monitor systems to deliver business value and to continually improve supporting processes and procedures
- **Performance Efficiency Pillar** — The performance efficiency pillar focuses on the structured and streamlined allocation of IT and computing resources. Key topics include selecting resource types and sizes optimized for workload requirements, monitoring performance, and maintaining efficiency as business needs evolve.
- **Reliability Pillar** — The reliability pillar focuses on workloads performing their intended functions and how to recover quickly from failure to meet demands. Key topics include distributed system design, recovery planning, and adapting to changing requirements.
- **Security Pillar** — The security pillar focuses on protecting information and systems. Key topics include confidentiality and integrity of data, managing user permissions, and establishing controls to detect security events.
- **Cost Optimization Pillar** — The cost optimization pillar focuses on avoiding unnecessary costs. Key topics include understanding spending over time and controlling fund allocation, selecting resources of the right type and quantity, and scaling to meet business needs without overspending.

- **Sustainability Pillar** — The sustainability pillar focuses on minimizing the environmental impacts of running cloud workloads. Key topics include a shared responsibility model for sustainability, understanding impact, and maximizing utilization to minimize required resources and reduce downstream impacts.

AWS Cloud Adoption Framework (AWS CAF)



Other Services

- **Amazon WorkSpace**
 - Managed Desktop as a Service (DaaS) solution to easily provision Windows or Linux desktops
 - Great to eliminate management of on-premise VDI (Virtual Desktop Infrastructure)
 - Fast and quickly scalable to thousands of users
 - Secured data – integrates with KMS
 - Pay-as-you-go service with monthly or hourly rates
- **Amazon AppStream 2.0**
 - Desktop Application Streaming Service
 - Deliver to any computer, without acquiring, provisioning infrastructure
 - The application is delivered from within a web browser

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- **AWS IoT Core**
 - AWS IoT Core allows you to easily connect IoT devices to the AWS Cloud
 - **Serverless, secure & scalable** to billions of devices and trillions of messages
 - Your applications can communicate with your devices even when they aren't connected
- **Amazon Elastic Transcoder**
 - ElasticTranscoder is used to **convert media files stored in S3 into media files in the formats required by consumer playback devices** (phones etc..)
- **AWS AppSync**
 - Store and sync data across mobile and web apps in real-time
 - **Makes use of GraphQL** (mobile technology from Facebook)
- **AWS Amplify** — A set of tools and services that helps you develop and deploy scalable full-stack web and mobile applications
- **AWS Device Farm** — Fully-managed service that tests your web and mobile apps against desktop browsers, real mobile devices, and tablets
- **AWS Backup** — Fully-managed service to centrally manage and automate backups across AWS services
- **Disaster Strategy**
 - Backup and Restore
 - Pilot Light
 - Warm Standby
 - Multi-site/Hot-site

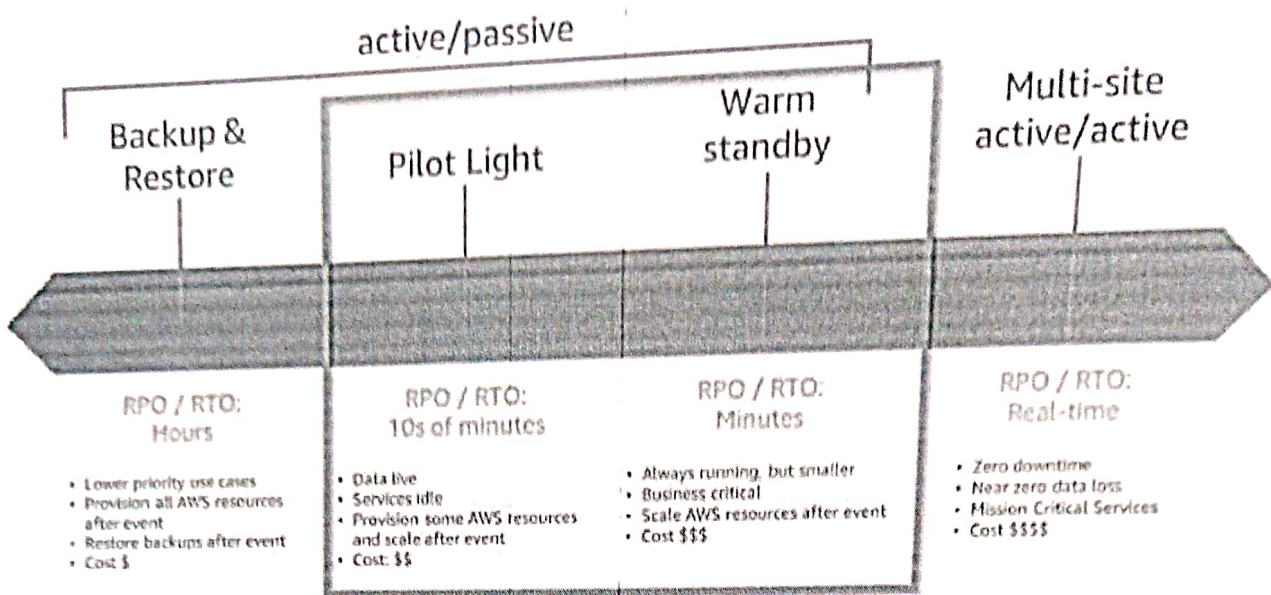


Figure 1. DR strategies

- **Amazon Elastic Disaster Recovery (DRS)** — Quickly and easily recover your physical, virtual, and cloud-based servers into AWS
- **AWS DataSync**
 - Move large amounts of data from on-premises to AWS
 - Replication tasks can be scheduled hourly, daily, weekly
 - The replication tasks are *incremental* after the first full load
- **AWS Application Discovery Service**
 - Plan migration projects by gathering information about on-premises data centers
 - Server utilization data and dependency mapping are important for migrations
 - **Agentless Discovery** (AWS Agentless Discovery Connector) → VM inventory, configuration, and performance history such as CPU, memory, and disk usage
 - **Agent-based Discovery** (AWS Application Discovery Agent) → System configuration, system performance, running processes, and details of the network connections between systems
 - Resulting data can be viewed within AWS Migration Hub
- **AWS Application Migration Service (MNG)**
- **AWS Migration Evaluator**
- **AWS Migration Hub**
 - Central location to collect servers and applications inventory data for the assessment, planning, and tracking of migrations to AWS

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- Helps accelerate your migration to AWS, automate lift-and-shift
 - **AWS Migration Hub Orchestrator** — provides pre-built templates to save time and effort migrating enterprise apps (e.g., SAP, Microsoft SQL Server...)
 - Supports migrations status updates from **Application Migration Service (MGN)** and **Database Migration Service (DMS)**

- **AWS Fault Injection Simulator (FIS)**

- A fully managed service for running fault injection experiments on AWS workloads
- Based on Chaos Engineering — stressing an application by creating disruptive events (e.g., a sudden increase in CPU or memory), observing how the system responds, and implementing improvements

- **AWS Step Functions**

- Build a serverless visual workflow to orchestrate your Lambda functions
- Use cases: order fulfillment, data processing, web applications, any workflow

- **AWS Ground Station**

- Fully managed service that lets you control satellite communications, process data, and scale your satellite operations
- Use cases: weather forecasting, surface imaging, communications, video broadcasts

- **Amazon Pinpoint**

- Scalable 2-ways (inbound/outbound) marketing communication service
- Supports email, SMS, push, voice, and in-app messaging
- Use cases: run campaigns by sending marketing, bulk, and transactional SMS messages